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Identifying changes in daily routine via passive sensing

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Summary

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- Changepoints Advertising
- Case study

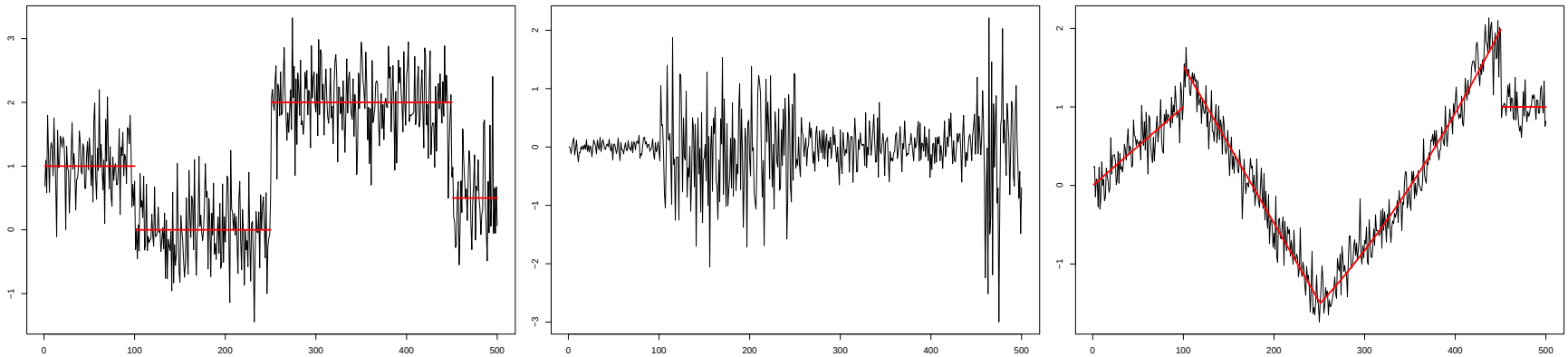


Changepoints Advertising

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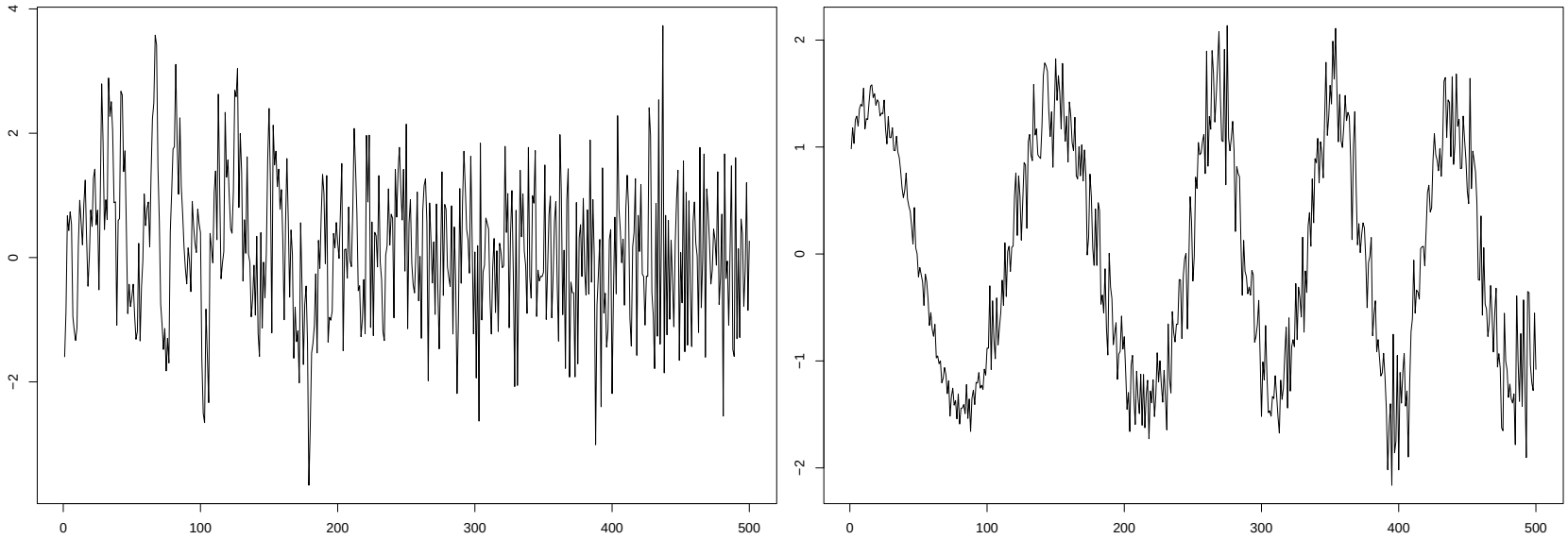
For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

There are many different types of change.



More realistic changes

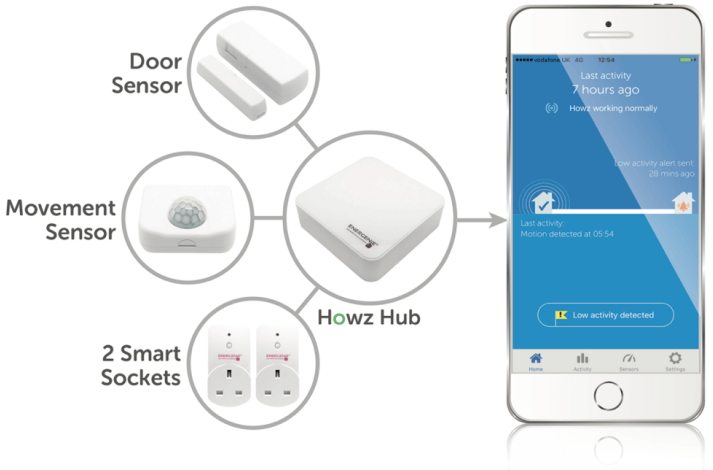
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Case Study: Howz

The Howz Smart System:

- Discreet passive sensors.
- Measure activity in the home.
- Learn about your daily routine.

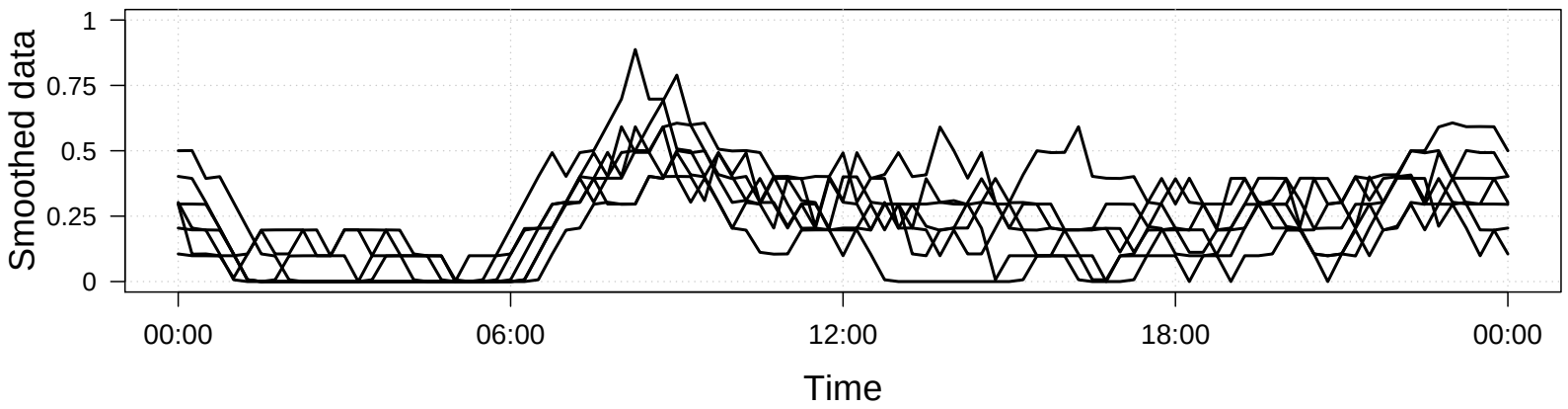


Objective:

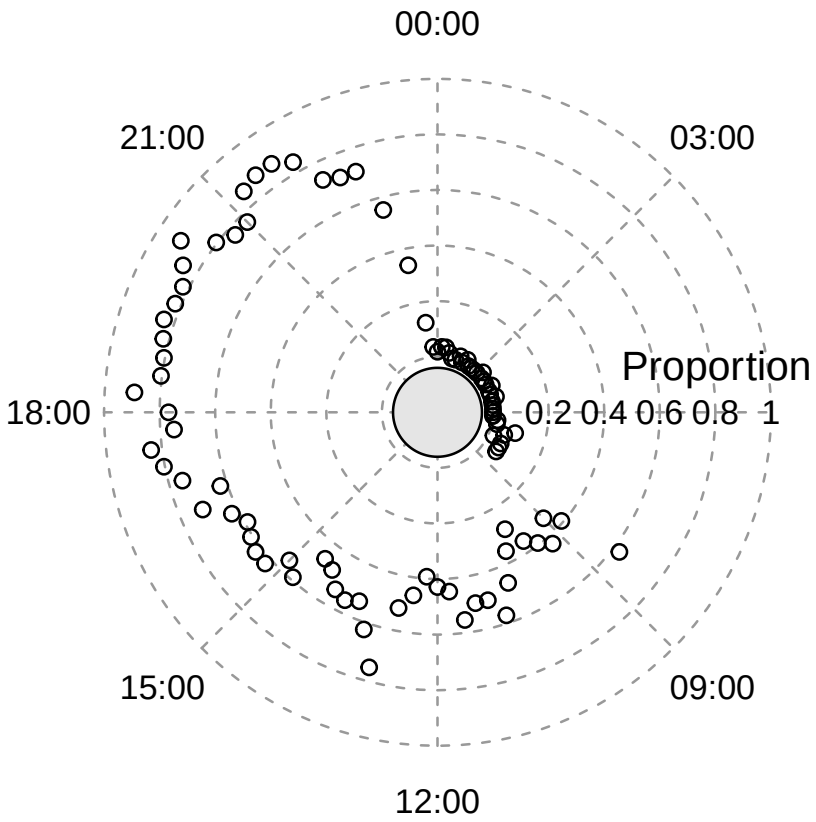
- Identify abnormal behaviour:
  - getting up later.
  - up in the night.
  - go out less frequently.
- Indicate a decline in health or well-being in an older person.

Data

- Data frequency at 15 minute intervals (96 obs. per day).
- Binary observations: activity detected (1) or none (0).
- Interest:
  - Transition times between different activities.
  - A measure of activity intensity.
- Time series presents a daily periodic pattern.

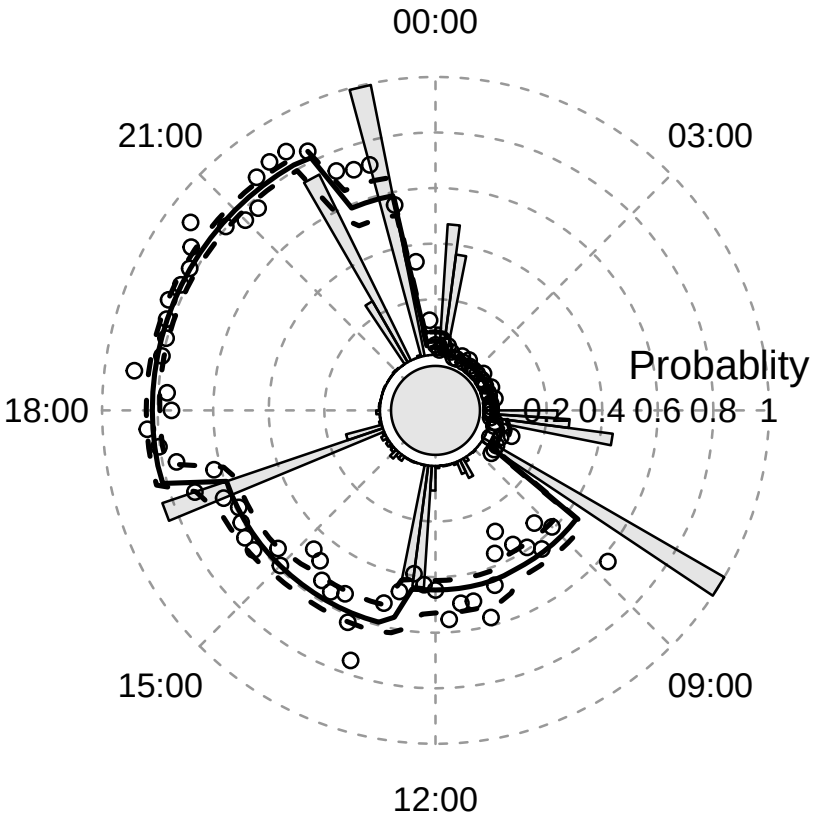


Case Study: Howz



- 8 weeks (Summer 2018)
- 15 minute intervals (N=96, n=5376)
- Assume $l = 4$ (1 hour)

Results



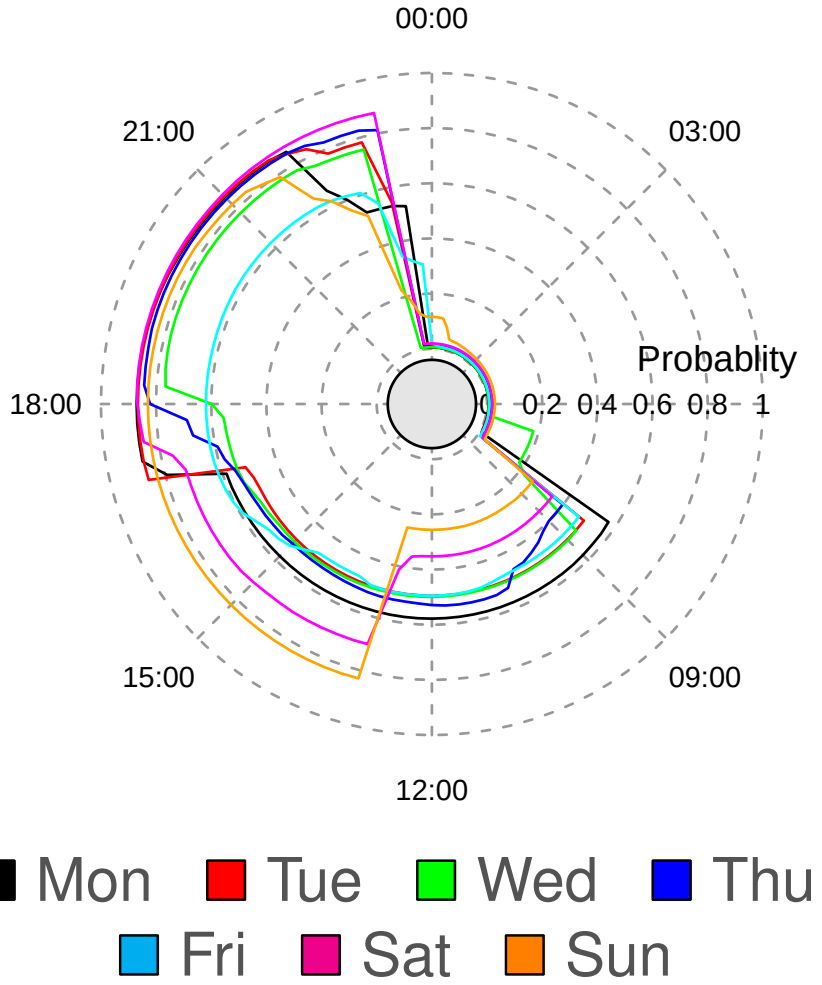
Modal number of segments:
$\hat{m} = 7$, $\hat{\pi}(\hat{m}|\boldsymbol{x}) = 75.1\%$

Modal changepoint vector, $\hat{\tau}$:

Time	$\hat{\pi}(i \in \tau \mid \boldsymbol{x})$	$\hat{\phi}_j$
00:30	47.1%	5.0%
06:45	44.3%	0.6%
08:15	100.0%	5.4%
12:45	41.5%	44.5%
16:45	83.8%	59.4%
22:15	74.6%	81.8%
23:15	99.4%	58.0%

$\hat{\pi}(\hat{\tau}|\boldsymbol{x}) = 5.2\%$
80% coverage: 268 events

Weekday effects



- Few events estimated:
  $\hat{m} = 4$ or 5.
- Consistent evening and night activity levels.
- Small variation in night and morning events.
- Weekday dependence on afternoon-evening transition:
  Weekday: 16:45–18:00.
  Weekend: 12:45.

Further developments

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- Have extended to identify changes in routines over time (paper under review)
- Are combining with other data

Other work:

- Changes in bone density as we age
- Characterizing curves to max CO2 for predicting fitness for surgery
- Developing statistics around passive (quantum) sensing of blood flow

- Want to know more . . . . . reach out for a chat
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